

The inverse sphere isometry gives the inverse radial map

MathlibAnnex.Sphere.radialExtension_leftInverse

theorem

Shows that extending the inverse sphere isometry radially undoes the forward radial extension.

Statement

Let X, Y be real normed spaces and $\Delta : S_X \rightarrow S_Y$ a bijective isometry for the ambient chord metrics. Here $S_X = \{u \in X : \|u\| = 1\}$ and $S_Y = \{v \in Y : \|v\| = 1\}$ are the unit spheres. Write $R_\Delta(0) = 0$ and $R_\Delta(x) = \|x\|\Delta(x/\|x\|)$ for $x \neq 0$, and define $R_{\Delta^{-1}}$ in the same way. Then

$$R_{\Delta^{-1}}(R_\Delta(x)) = x \quad (x \in X).$$

Assumptions

The only geometric assumptions are that X, Y are real normed spaces and Δ is a bijective isometry between their unit spheres. The inverse used here is the inverse of that same Δ . Completeness, finite-dimensionality and a positive dimension are not required.

Conclusion

The map $R_{\Delta^{-1}}$ is a left inverse of R_Δ . Applying this result to Δ^{-1} gives $R_\Delta(R_{\Delta^{-1}}(y)) = y$ for all $y \in Y$, recorded separately as the right-inverse theorem. The two radial maps therefore give a bijection of the ambient spaces.

Bundling these two maps as `radialExtensionEquiv` uses precisely these inverse identities. The later homeomorphism uses the same maps with continuity proofs. Neither bundling chooses a new map.

Proof route

For $x = 0$ both maps give zero. For $x \neq 0$, set $r = \|x\| > 0$, $u = x/r$ and $y = R_\Delta(x) = r\Delta(u)$. Since $\Delta(u) \in S_Y$, norm preservation gives $\|y\| = r > 0$, so $y \neq 0$ and $y/\|y\| = \Delta(u)$. Substitution into the inverse radial formula gives

$$R_{\Delta^{-1}}(y) = r\Delta^{-1}(\Delta(u)) = ru = x.$$

The cancellation uses the inverse of the given sphere equivalence, and positive normalization makes the argument of that inverse exactly $\Delta(u)$.

▼ Proof steps

1. For $x \neq 0$, the norm identity rules out $R_\Delta(x) = 0$. Thus the nonzero branch of the inverse radial definition is applicable.
2. Normalize $r\Delta(u)$ using $r > 0$ and $\|\Delta(u)\| = 1$. Its unit direction is $\Delta(u)$, so the inverse sphere map returns u , and multiplication by the preserved radius returns x .

Main citations

- Exact declaration and its source — `MathlibAnnex.Sphere.radialExtension_leftInverse`
- The radial map and its full definition — `MathlibAnnex.Sphere.radialExtension`
- Preservation of the radius — `MathlibAnnex.Sphere.radialExtension_norm`
- The opposite composition — `MathlibAnnex.Sphere.radialExtension_rightInverse`
- The same maps bundled as an equivalence — `MathlibAnnex.Sphere.radialExtensionEquiv`

Lean source signature (exact)

```
theorem radialExtension_leftInverse
  (e : Metric.sphere (0 : X) 1 ≈i Metric.sphere (0 : Y) 1) :
  LeftInverse (radialExtension e.symm) (radialExtension e)
```

In the source	Mathematical meaning
<code>(e : Metric.sphere (0 : X) 1 ≈_i Metric.sphere (0 : Y) 1)</code>	The same bijective chord-distance isometry $\Delta : S_X \rightarrow S_Y$, where $S_X = \{x : \ x\ _X = 1\}$ and $S_Y = \{y : \ y\ _Y = 1\}$. The surrounding binders make X, Y real normed spaces.
<code>e.symm</code>	The inverse $\Delta^{-1} : S_Y \rightarrow S_X$ of that same sphere isometry.
In the source	Mathematical meaning
<code>radialExtension e; radialExtension e.symm</code>	The radial maps $R_\Delta : X \rightarrow Y$ and $R_{\Delta^{-1}} : Y \rightarrow X$ defined in this Card by their zero and radius-direction formulas.
<code>LeftInverse (radialExtension e.symm) (radialExtension e)</code>	For every $x \in X$, $R_{\Delta^{-1}}(R_\Delta(x)) = x$. The order of the maps is part of the conclusion.

▼ Exact surrounding binder context (separate excerpt)

```
namespace MathlibAnnex
namespace Sphere

universe u v

variable {X : Type u} {Y : Type v}
[NormedAddCommGroup X] [NormedSpace ℝ X]
[NormedAddCommGroup Y] [NormedSpace ℝ Y]
```

▼ **Exact content identity**

Declaration: `MathlibAnnex.Sphere.radialExtension_leftInverse`

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